

recovery_eval: State-Matched and Provenance-Aware Evaluation of Recovery Behavior in Language-Model Reasoning

Sham Satish Thakare
Independent Researcher
Pune, Maharashtra, India
Email: shamthakare3000@gmail.com

Abstract—Evaluating whether different language-model policy configurations exhibit improved recovery from intermediate reasoning errors is difficult because overall continuation quality can confound recovery-specific behavior. We introduce `recovery_eval`, a reproducible data-centric evaluation framework that compares verifier-defined recovery states with prospectively matched reference-control states using frozen structural covariates. The framework combines state matching with append-only exposure governance, primitive neural-rollout provenance, and independent analysis reconstruction. We apply `recovery_eval` to 400 genuine neural continuations generated by two released Qwen2.5-Math 1.5B checkpoint-interface configurations on 20 prospectively isolated GSM8K evaluation problems. The Instruct checkpoint showed higher observed continuation success than the Base checkpoint in both recovery and matched-control states, but the improvement was larger for controls. The resulting matched recovery-specific contrast was minus 0.110, with a descriptive 95 percent problem-level bootstrap interval from minus 0.240 to 0.030. Thus, under this evaluation, we did not observe evidence of a recovery-specific advantage for the Instruct checkpoint. The result illustrates how aggregate benchmark improvements can obscure state-specific reasoning behavior and motivates provenance-aware, state-matched evaluation for language-model diagnostics.

Index Terms—Large Language Models, Mathematical Reasoning, Model Evaluation, Data-Centric AI, Reproducibility

I. Introduction

Large language models (LLMs) trained on multi-step mathematical reasoning demonstrate remarkable capacity for generating complex chain-of-thought solutions [1], [2]. A foundational question in AI evaluation is whether different released language-model configurations exhibit recovery behavior beyond their general continuation-quality differences [3]–[5].

Evaluating error-recovery capability is fraught with statistical confounding. Standard benchmark evaluation measures aggregate end-to-end accuracy, which conflates baseline generation fluency with true error recovery. Simply comparing model rollouts from error-containing prefixes against valid prefixes ignores critical confounding covariates, including trajectory depth, remaining solution complexity, problem difficulty, and token length. A post-trained model may exhibit higher accuracy on error prefixes simply because its overall baseline generation

quality has improved across all prompt conditions, rather than possessing a specialized error-recovery mechanism [6], [7].

To address these diagnostic challenges, we introduce `recovery_eval`, a data-centric evaluation infrastructure for measuring state-specific recovery behavior in reasoning LLMs. The primary contributions of this work are three-fold:

- 1) **Verifier-Defined State Matching Protocol:** A prospective matching methodology that pairs verifier-identified error states with valid reference-control states using frozen structural covariates and an exact matching constraint on categorical problem attributes.
- 2) **Provenance and Exposure Governance Architecture:** An append-only evidence ledger that locks evaluation items, logs primitive BPE token-level generation continuations, verifies model weight manifests, and enables 100% independent analytical reconstruction.
- 3) **Genuine Two-Checkpoint Framework Demonstration:** A 400-rollout empirical evaluation across Qwen2.5-Math-1.5B Base and Instruct checkpoints, demonstrating that aggregate continuation success gains (+0.4300 on recovery vs +0.5400 on control) do not automatically translate into a detectable recovery-specific advantage ($D_{\text{recovery}} = -0.1100$).

Figure 1: `recovery_eval` End-to-End Governance & Evaluation Pipeline



Fig. 1. `recovery_eval` End-to-End Governance & Evaluation Pipeline.

II. Related Work

Prior work has studied end-to-end reasoning accuracy, process supervision, self-correction, and search-based reasoning [1], [3]–[10]. Our framework instead focuses on prospective structural matching between verifier-defined recovery and control states. In the primary-source corpus we audited, we did not identify prior work combining verifier-defined recovery states, prospective structural matching, exposure governance, and primitive rollout provenance within a single evaluation protocol.

TABLE I
Comparison with Existing Reasoning Evaluation Frameworks

Framework / Work	Matching	Ledger	Provenance
End-to-End GSM8K [1]	None	No	Final Answer
PRM800K [3]	Unmatched	Partial	Step Scores
STaR [4]	Unmatched	No	Filtered Traces
Tree of Thoughts [10]	Unmatched	No	Node Search
recovery_eval (Ours)	Matched	Append-Only	Full BPE Token

Existing benchmark designs generally do not explicitly control for the structural state differences targeted by our matched recovery/control protocol.

III. Problem Formulation

Let s represent a partial reasoning trajectory prefix for a mathematical word problem $p \in \mathcal{P}$. A deterministic verifier $\mathcal{V}(s) \in \{0, 1\}$ evaluates whether prefix s contains any logical or arithmetic errors.

A recovery state s_R is a partial prefix containing a verifier-identified intermediate error. A reference-control state s_C is a valid intermediate prefix for the same problem instance at an equivalent stage of resolution.

For a target policy π and baseline policy π_0 , let $V(s) \in \{0, 1\}$ denote the binary continuation success indicator of completing the solution correctly from prefix s . We define the matched recovery-specific contrast D_{recovery} as:

$$D_{\text{recovery}} = \mathbb{E}[V_{\pi}(s_R) - V_{\pi_0}(s_R)] - \mathbb{E}[V_{\pi}(s_C) - V_{\pi_0}(s_C)] \quad (1)$$

IV. The recovery_eval Framework

The recovery_eval framework provides an end-to-end evaluation pipeline comprising three main modules: state generation, prospective matching, and governance-backed neural rollout execution.

A. System Architecture

The framework operates as a decoupled pipeline: 1) Registry Phase: Evaluation items and candidate prefixes are locked in an append-only registry; 2) Matching Phase: Recovery states are prospectively matched to reference control states prior to inference; 3) Execution Phase: Continuations are sampled under sealed generation settings and evaluated by the primitive verifier.

V. State Construction and Prospective Matching

Recovery states s_R are constructed by introducing controlled single-step arithmetic perturbations into valid intermediate reasoning prefixes. For example, if a valid intermediate calculation step evaluates to $11 \times 4 = 44$, a recovery perturbation injects a single arithmetic error $11 \times 4 = 46$ while preserving all prior problem context and equations. Reference control states s_C maintain valid prefix steps.

Figure 2: Verifier-Defined Recovery State vs Matched Reference Control State Construction

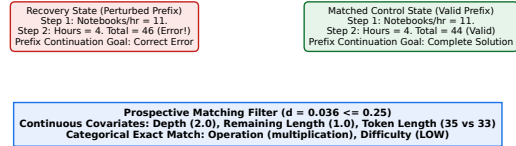


Fig. 2. Verifier-Defined Recovery State vs Matched Reference Control State Construction.

To prevent confounding due to trajectory length or problem complexity, each recovery state s_R is matched to a control state s_C using a normalized weighted-L1 Manhattan distance over continuous structural covariates:

$$d(i, j) = \sum_{k=1}^K w_k \frac{|x_{ik} - x_{jk}|}{s_k} \quad (2)$$

subject to an exact categorical matching constraint on reasoning operation type and problem difficulty. Table II lists the prospective matching parameters. Matching is used here as a descriptive design tool to improve structural comparability between recovery and control states; it is not used to identify a causal treatment effect [11], [12].

TABLE II
Frozen Prospective Matching Covariates and Scales

Covariate Name	Type	Weight (w_k)	Scale (s_k)
Trajectory Depth	Continuous	0.4	1.5
Remaining Solution Length	Continuous	0.4	1.0
Token Length	Continuous	0.2	15.0
Reasoning Operation	Categorical	Exact Match	–
Problem Difficulty	Categorical	Exact Match	–

VI. Provenance and Exposure Governance

To guarantee artifact integrity and prevent item re-selection or data leakage, recovery_eval implements an append-only event ledger [13]. Every item exposure event is logged with cryptographic parent hashing. Development-exposed items are flagged as SIMULATION_EXPOSED and forbidden from entering the evaluation registry.

Figure 3: Append-Only Immutable Primitive Evidence Provenance Chain

PREECUTION LOCK SHA-256: d3b14589...	REAL model.generate() PyTorch Output Tensor	BPE DECODE Assert Token Round-trip	RAW JSONL PROVEN SHA-256: 51b5a157d9e44102caeb86d0...	INDEPENDENT RECONSTRUCTION ES Contrast: -0.110
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Fig. 3. Append-Only Immutable Primitive Evidence Provenance Chain.

Each raw rollout record in RAW_NEURAL_ROLLOUTS.jsonl captures:

- Input token IDs and generated BPE token continuation IDs.
- Monotonic generation timestamps in UTC+05:30.
- Resolved HuggingFace repository commit hashes and local weight manifest SHA-256 digests.
- Deterministic verifier execution outputs.

VII. Experimental Design

We evaluate the framework across 20 fresh, prospectively isolated GSM8K test items ($N = 20$). For each problem item, 1 recovery state and 1 matched control state are evaluated across 2 policy configurations (Qwen2.5-Math-1.5B Base and Instruct) using 5 stochastic rollout seeds ($S = 5$), yielding exactly 400 continuations ($20 \times 2 \times 2 \times 5 = 400$).

Model continuations were sampled at temperature $t = 0.0$ (greedy decoding) on an Apple Silicon MPS device (mps:0) using PyTorch model.generate(). Table III details the execution design and provenance metrics.

TABLE III
Evaluation Design and Provenance Statistics

Attribute	Specification / Value
Base Model	Qwen2.5-Math-1.5B (4a83ca6e)
Instruct Model	Qwen2.5-Math-1.5B-Instruct (aafef0fc)
Hardware Device	Apple Silicon MPS (mps:0)
Total Rollouts	400 (200 Base, 200 Instruct)
Total Tokens	19,212 BPE tokens
Measured Duration	1,755.86 seconds (≈ 29.26 min)
Base Throughput	11.86 tok/s (11,354 tokens / 957.37s)
Instruct Throughput	9.84 tok/s (7,858 tokens / 798.49s)
BPE Decode Match	100.0% (400/400 exact match)
Raw Record SHA	51b5a157d9e44102caeb86d0...

VIII. Results

Table IV presents continuation success rates across states and policy configurations.

Under the evaluated state-matched protocol, we did not observe evidence of a recovery-specific advantage for the Instruct checkpoint over the Base checkpoint. The

TABLE IV
Observed Empirical Continuation Outcomes and Contrast

State Condition	Base (π_0)	Instruct (π)	Difference (Δ)
Recovery States (s_R)	0.1500	0.5800	+0.4300
Control States (s_C)	0.3800	0.9200	+0.5400
Matched Contrast (D_{rec})	–	–	–0.1100

estimated matched recovery-specific checkpoint-interface contrast was -0.1100 , with a 95% descriptive problem-level bootstrap interval of $[-0.240, +0.030]$ (10,000 resamples).

Figure 4: Observed Recovery/Control Differences & Matched Contrast (N=400)

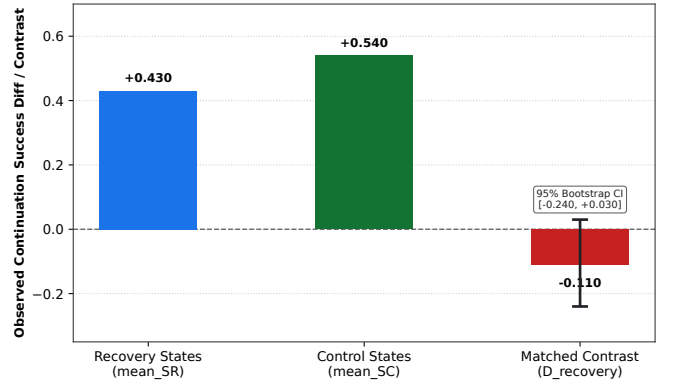


Fig. 4. Observed Recovery/Control Policy Differences and Matched Recovery Contrast (N=400).

The Instruct checkpoint showed higher continuation success than Base in both recovery (+0.4300) and control (+0.5400) states, but the improvement was larger for controls. Consequently, aggregate benchmark gains do not automatically translate into a recovery-specific advantage.

IX. Sensitivity and Reproducibility Analysis

We assess matching quality across the 20 matched pairs. The mean normalized weighted-L1 distance is $d_{\text{mean}} = 0.0360$, median is $d_{\text{median}} = 0.0360$, and maximum distance is $d_{\text{max}} = 0.0360$. All 20 pairs (20/20) satisfy both the standard ($d \leq 0.25$) and tight ($d \leq 0.10$) thresholds [14].

Table V reports covariate balance metrics constrained strictly to single column width. Standardized Mean Differences (SMDs) for trajectory depth and remaining length are +0.0000, while token length SMD is +0.1333.

Independent re-computation from sealed raw evidence RAW_NEURAL_ROLLOUTS.jsonl (SHA-256: 51b5a157d9e44102caeb86d0...) yields exact agreement (0.0 discrepancy across all metrics).

TABLE V
Matching Distance and Covariate Balance Metrics

Metric / Covariate	Raw Diff	SMD / Distance
Trajectory Depth	+0.0000	+0.0000
Remaining Solution Length	+0.0000	+0.0000
Token Length	+2.0000	+0.1333
Mean Distance (d_{mean})	–	0.0360 ($20/20 \leq 0.25$)
Max Distance (d_{max})	–	0.0360 ($20/20 \leq 0.10$)

X. Limitations

This study has explicit methodological scope boundaries:

- **Model Scope:** Evaluated on a single model family (Qwen2.5-Math-1.5B) across two released checkpoints.
- **Interface Scope:** Model weight states and intended prompt interfaces differ between Base and Instruct checkpoints.
- **Benchmark Scope:** Evaluation uses 20 GSM8K test items ($N = 20$). Benchmark pretraining contamination cannot be excluded.
- **Sample Size:** 5 stochastic rollout continuations per state/policy represent evaluation samples, not independent model training replications.
- **No Causal Claim:** Results represent a descriptive contrast under state matching, not a causal claim regarding post-training mechanisms.

XI. Conclusion

We presented `recovery_eval`, a prospective state-matched evaluation framework for language-model reasoning diagnostics. By pairing verifier-defined recovery states with structural reference controls and maintaining primitive rollout governance, the framework enables fine-grained evaluation of reasoning policies without confounding baseline fluency gains.

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